Generative model: categorical states, observations, actions, and hierarchy

The 9 studies — including the contaminated-sentinel robustness sweep (Study 4) — run on one shared sentinel world (Studies 5–7 use its moving and hierarchical variants):

Generative model (part 2)

a discrete sentinel partially-observable Markov decision process (POMDP) using the categorical world structure illustrated by Friston et al. (2024), Figures 1 and 4.

Generative model (part 3)

We adopt the discrete-state active-inference formulation that the community has standardized around — the categorical $A/B/C/D_0$ generative model of da Costa et al. (2020), the same object the `pymdp` (Heins et al. 2022) and RxInfer (Bagaev et al. 2023) toolboxes operate on.

Generative model (part 4)

We reimplement it in pure NumPy in `pomdp.py` so the colony, its sensors, and its dynamics are exactly the ones the analysis executes.

A colony of sentinels watches a single hidden creature whose location is the shared latent factor they federate beliefs about.

Generative model (part 5)

The structural map in fig. 7 follows the same generative-model vocabulary while making the implementation boundary visible: Panel A shows what one agent actually sees — one noisy categorical report over the 9 possible locations —

Generative model (part 6)

while Panel B identifies the $A/B/C/D_0$ factors that turn that report into a local posterior.

Generative model (part 7)

Temporal depth then describes state, observation, posterior, and control order; hierarchical depth describes conditioned priors. It is a formal schematic, not an assertion that every displayed dependency is simultaneously estimated in every study.

Generative model (part 8)

Hidden location s

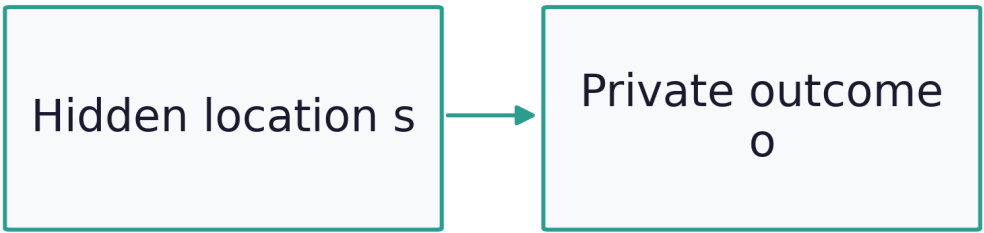


Generative model (part 9)

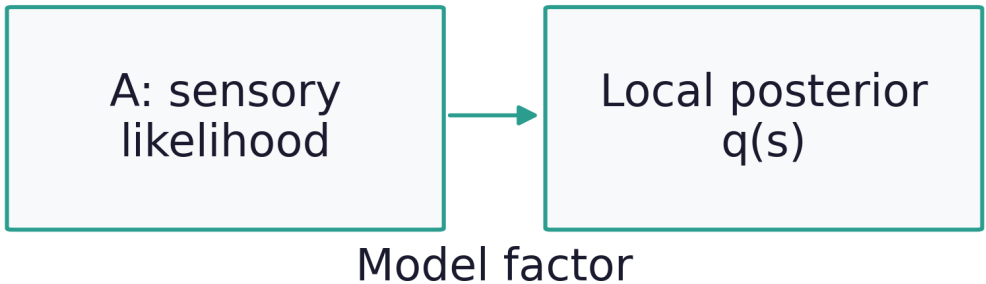
Private outcome o



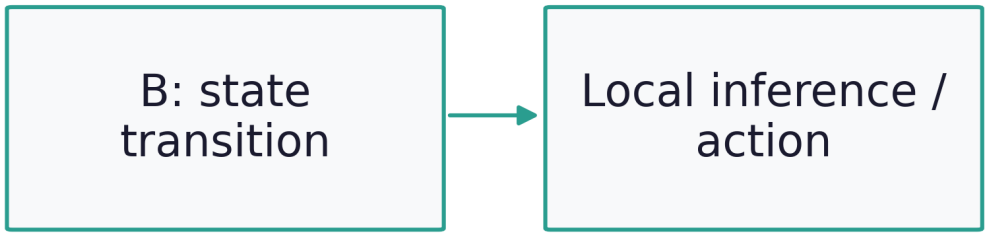
Generative model (part 10)



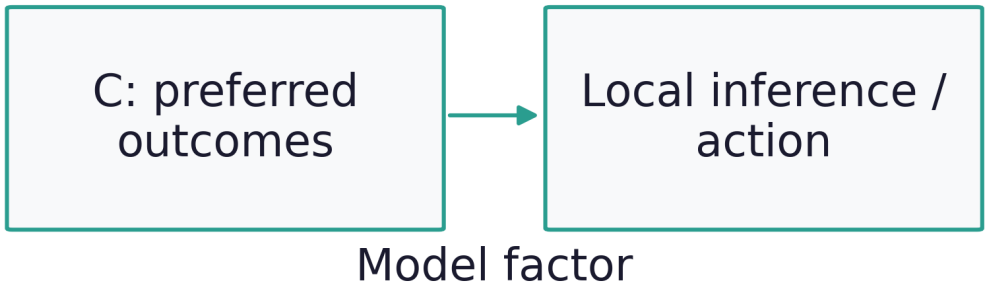
Generative model (part 11)



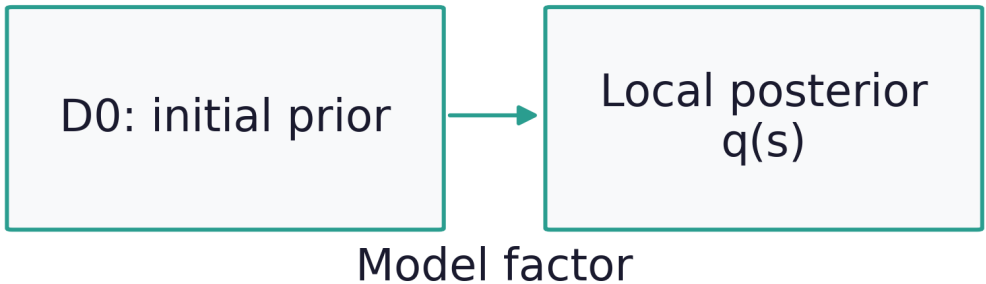
Generative model (part 12)



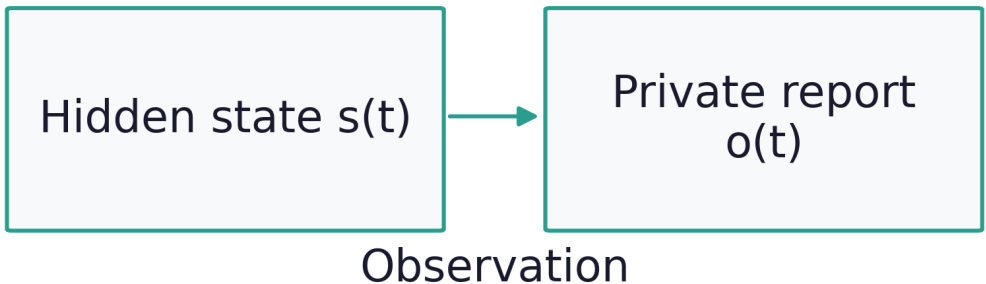
Generative model (part 13)



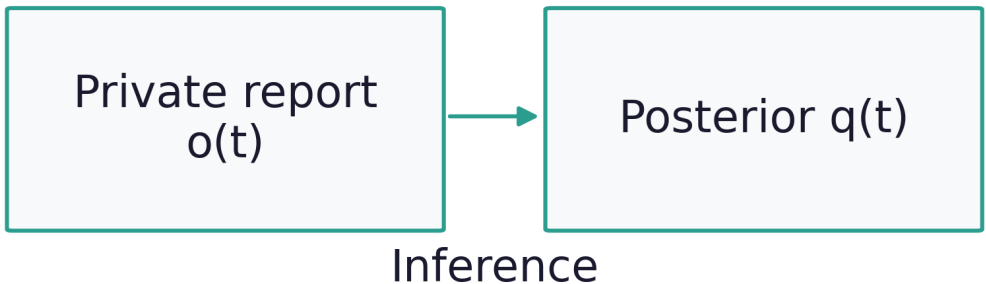
Generative model (part 14)



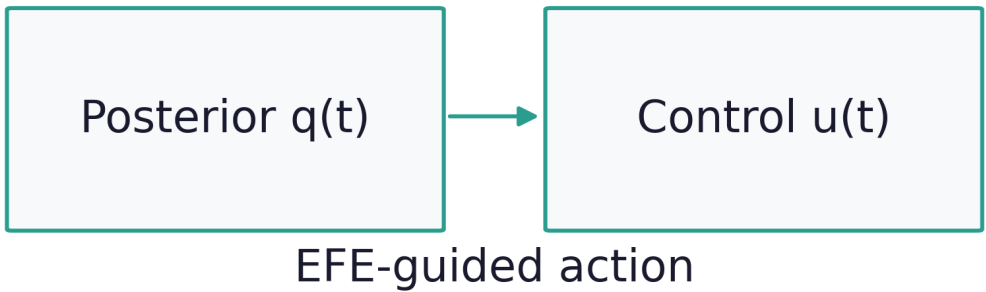
Generative model (part 15)



Generative model (part 16)

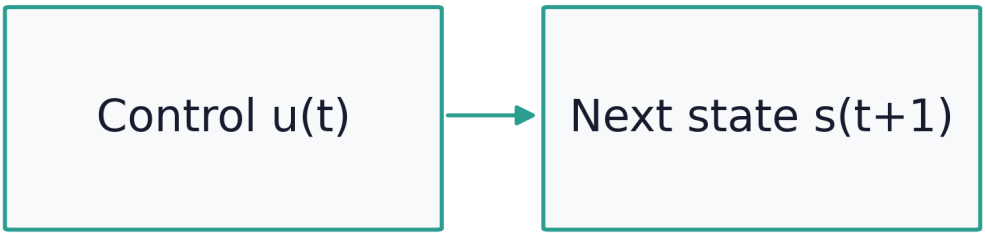


Generative model (part 17)

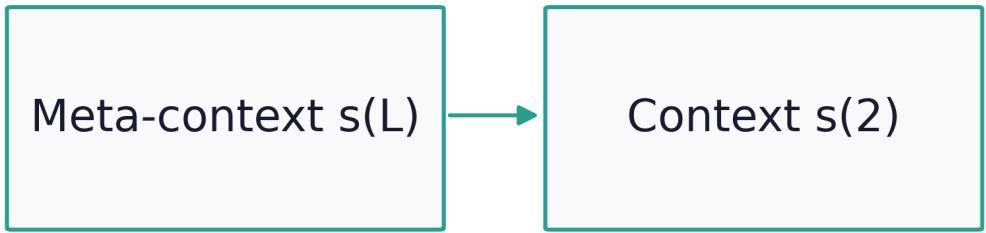


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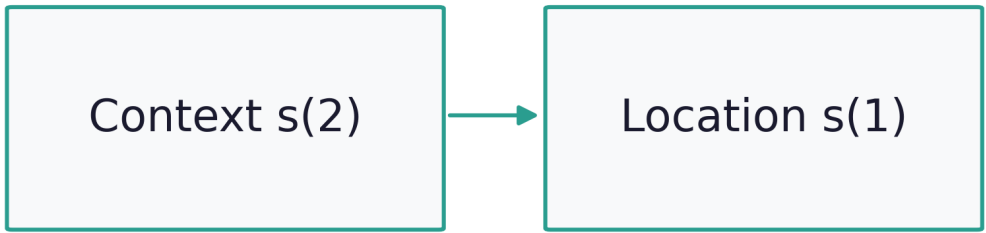
Generative model (part 18)



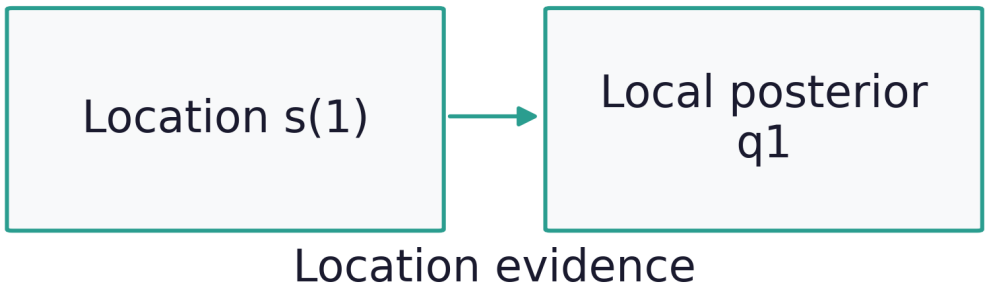
Generative model (part 19)



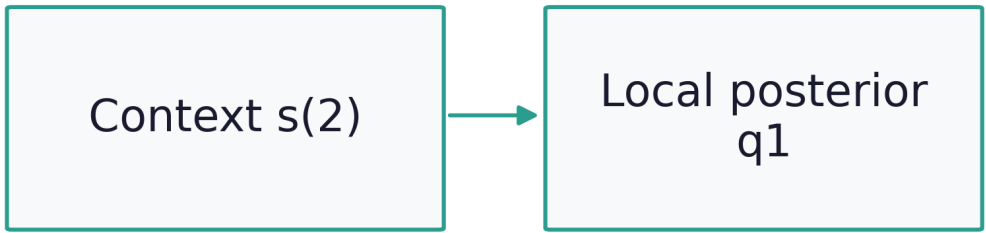
Generative model (part 20)



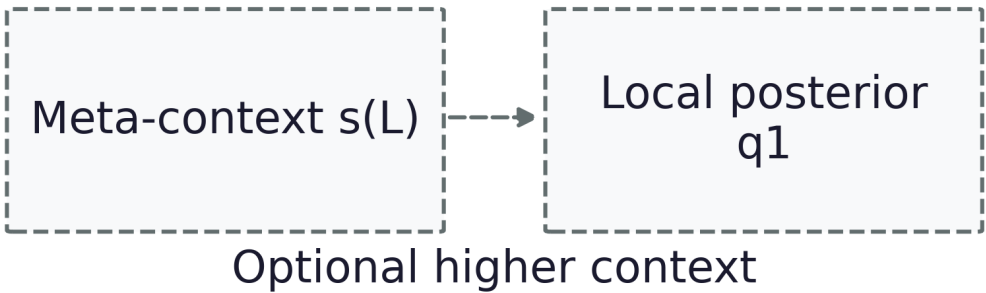
Generative model (part 21)



Generative model (part 22)



Generative model (part 23)



Generative model (part 24)

Diagonal mass is sensor acuity; residual mass is categorical noise. Raw outcomes remain private; only posteriors are broadcast.

Generative model (part 25)

$$q(s) = \mathrm{softmax}(\ln D_0 + \ln A[o, \cdot])$$

Generative model (part 26)

Project recovery identity: robust_aggregate(0) = log_linear_pool. This limit does not reconstruct the full source protocol.

Generative model (part 27)

$$\bar{q}_1 = \sum_k q_2[k] D_{1|k}$$

Generative model (part 28)

Flat studies stop after posterior sharing. Moving-world studies execute B and EFE-guided control. Hierarchy is an optional extension.

Generative model (part 29)

Schematic formalization, not an empirical estimate. Private sensory outcomes become posterior messages; fusion uses the shared categorical state.

State space: one shared latent factor

The world holds one hidden factor: the creature's location on a square grid of side L, giving $n_s = L^2$ location states. Our sentinel world uses the 3×3 cardinality illustrated in Friston et al. (2024), Fig. 1, so $n_s = 9$ — the cardinality `pomdp.N_LOCATIONS` exposes and `experiment_config` carries as `n_locations`.

State space: one shared… (part 2)

This single location factor is precisely the latent the colony gossips about: it is the shared argument of the log-linear pool (eq. 9) and of every belief-sharing round (eq. 14). Fixing one hidden factor keeps the recovery limits of sec. 9 closed-form and exactly testable,

State space: one shared... (part 3)

rather than approximated.

Four categorical tensors: likelihood, transitions, preferences, priors

The generative model is the tuple (A, B, C, D_0) in the discrete active-inference convention: a categorical probability mass function is a non-negative vector summing to one,

Four categorical tensors (part 2)

and a likelihood matrix is shape (n_o, n_s) whose columns (indexed by hidden state) are categorical.

Four categorical tensors (part 3)

Observation likelihood $A = P(o \mid s)$**.** Each sentinel observes the creature's cell through a noisy sensor. With probability `acuity` the sensor reports the true cell; the residual mass $1 -$ acuity spreads uniformly over the other $n_s - 1$ cells. With outcome cardinality $n_o = n_s$,

Four categorical tensors (part 4)

the single location modality is one (n_s, n_s) matrix:

Four categorical tensors (part 5)

$$
\begin{aligned}
A_{os} &= \text{acuity} && \text{if } o = s, \\
A_{os} &= \frac{1 - \text{acuity}}{n_s - 1} && \text{if } o \neq s, \\
\sum_o A_{os} &= 1.
\end{aligned}
\tag{15}
$$

Four categorical tensors (part 6)

The acuity in eq. 15 tunes how peaked the sensor is: high acuity gives a near-diagonal A that pins the creature; the belief-sharing study deliberately runs the colony at the low acuity acuity $= 0.55$, where no single sentinel can resolve the location alone and the colony must pool evidence to do so.

Four categorical tensors (part 7)

When a seeded generator is supplied, each sentinel’s acuity is jittered by a small non-negative perturbation, so a colony carries slightly heterogeneous likelihoods while every column remains a proper pmf.

Four categorical tensors (part 8)

Transition tensor $B = P(s' \mid s, u)$**.** The creature moves on the grid under three control paths — `still`, `left`, `right` — so B has shape (n_s, n_s, n_u) with $n_u = 3$. `still` is the deterministic self-loop; `left` and `right` decrement and increment the grid column,

Four categorical tensors (part 9)

saturating at the walls (a wall-adjacent move in the wall's direction is a self-loop).

Four categorical tensors (part 10)

All three controls act on the column index alone, so the creature's row is preserved and its motion is confined to the horizontal axis of the grid — a deliberate one-dimensional control over the two-dimensional location factor. Every slice $B_{\cdot\cdot u}$ is column-normalized by construction,

Four categorical tensors (part 11)

so the transition is a valid categorical for each action.

Four categorical tensors (part 12)

Log-preference C. The sentinel prefers to *see* the creature near the den (the center cell), encoded as a log-preference vector of shape $(n_o, 1)$ with a positive bump on the center outcome and zero elsewhere.

Four categorical tensors (part 13)

The preferred-outcome distribution that the expected-free-energy decomposition of sec. 8.15 uses is $p_C(o) = \mathrm{softmax}(C)[o]$.

Four categorical tensors (part 14)

Initial prior D_0. The creature is believed to start at the grid center, so D_0 of shape $(n_s, 1)$ places unit mass on the center cell. D_0 enters state inference as the log-prior of the one-step variational update (eq. 16 below).

Four categorical tensors (part 15)

The columns-are-pmfs invariant is not assumed — it is pinned by ISC-15, which checks that every column of A and of each $B_{\cdot\cdot u}$ sums to one.

One-step variational state inference in the grid world

Given an observation o, a sentinel forms a posterior over the creature's location by a single softmax step (Friston et al. (2024), Eq. 4): the log-prior plus the additive log-likelihood message,

One-step variational state... (part 2)

summed over any conditionally independent modalities m,

One-step variational state... (part 3)

$$q(s) = \mathrm{softmax}\left(\begin{array}{l} \ln D_0(s) \\ \quad + \sum_m \ln A_m[o_m, s] \end{array} \right). \tag{16}$$

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One-step variational state... (part 4)

The message $\ln A_m[o_m, \cdot]$ is the row of A_m that the observed outcome selects; summing messages over modalities makes each modality an additive evidence term — the categorical product-of-experts. The companion variational free energy, the scalar eq. 16 minimizes, is

One-step variational state... (part 5)

$$
\begin{aligned}
F[q] &= \mathbb{E}_q\left[\begin{array}{c}\ln q(s) - \ln D_0(s) \\ -\sum_m \ln A_m[o_m, s]\end{array}\right] \\
&= \mathrm{KL}(q \,\|\, D_0) - \mathbb{E}_q\big[\textstyle\sum_m \ln A_m[o_m, s]\big].
\end{aligned}
\tag{17}
$$

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One-step variational state... (part 6)

reported in nats. The one-step posterior of eq. 16 is its unique minimizer, where F equals the negative log model evidence.

One-step variational state... (part 7)

Both live in `belief_updating.infer_states` and `belief_updating.vfe`, and the free energy of eq. 17 is the quantity the communicating-versus-incommunicado colony comparison of sec. 8.22 scores.

One-step variational state... (part 8)

This inference step is not a separate mechanism bolted onto the colony: it is the $L = \mathrm{NLL}$, learning-rate-1 special case of the generalized-Bayes posterior eq. 2, reusing the same locked softmax.

One-step variational state... (part 9)

That client identity recovers the stated categorical Bayes substrate at its trusting limits. The separate server identity in sec. 8.8 then yields the project log-linear pool under its qualified Eq. 7 message-combination bridge;

One-step variational state... (part 10)

together these do not recover the complete Friston protocol (sec. 9).

Hidden-state to action loop: the POMDP substrate

The categorical POMDP loop in fig. 8 separates the common latent-state substrate from the federation transport. In the sentinel interpretation, an agent is a location-sensitive observer:

Hidden-state to action loop (part 2)

the hidden state is one of 9 cells, the private outcome is a noisy categorical report of that location, and the agent sends its posterior over the location rather than the report itself.

Hidden-state to action loop (part 3)

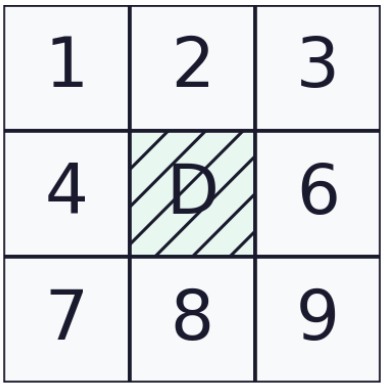
The flat belief-sharing studies use the observation, posterior, and communication subset; the moving-world extension also executes transition and EFE-guided action paths.

Hidden-state to action loop (part 4)

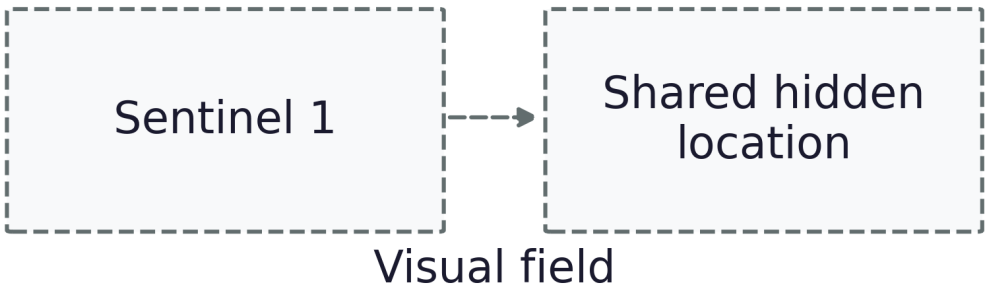
The diagram therefore gives readers the active-inference context without turning a conceptual loop into a claim that every study estimates every latent or policy quantity.

Hidden-state to action loop (part 5)

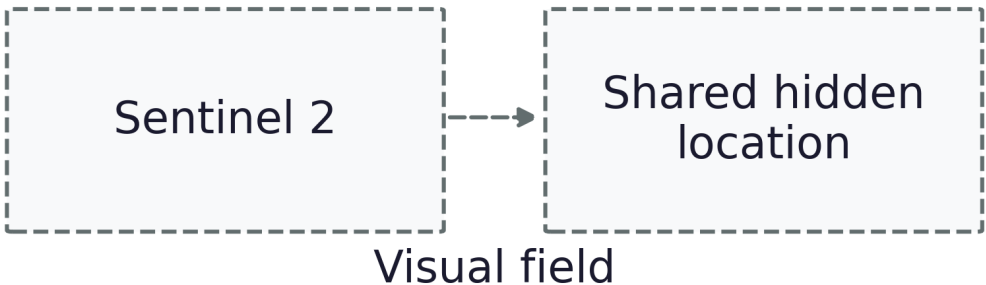
Shared world; D = den



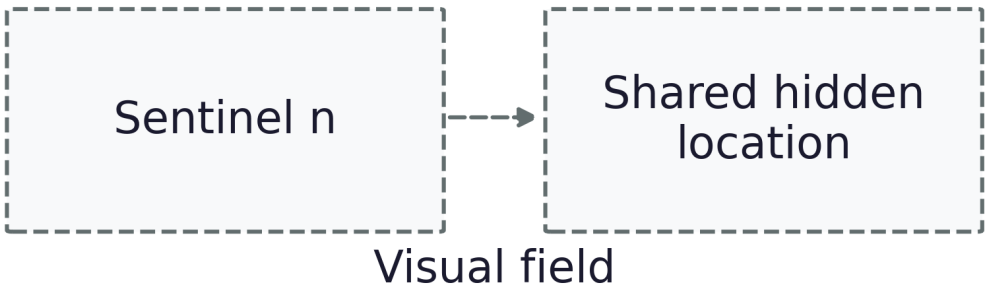
Hidden-state to action loop (part 6)



Hidden-state to action loop (part 7)



Hidden-state to action loop (part 8)



Hidden-state to action loop (part 9)

Schematic local posterior q1



Hidden-state to action loop (part 10)

Schematic local posterior q2

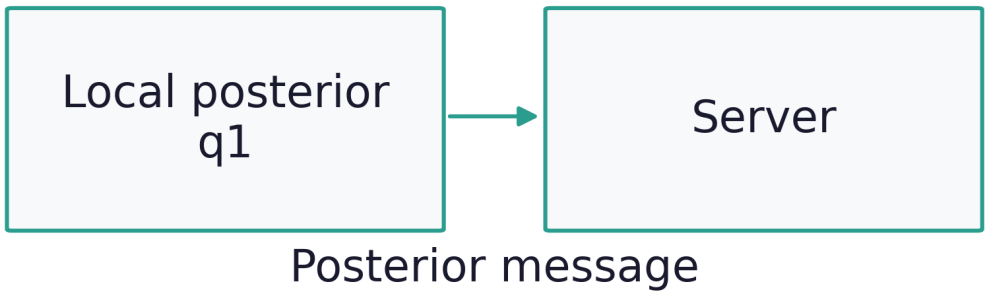


Hidden-state to action loop (part 11)

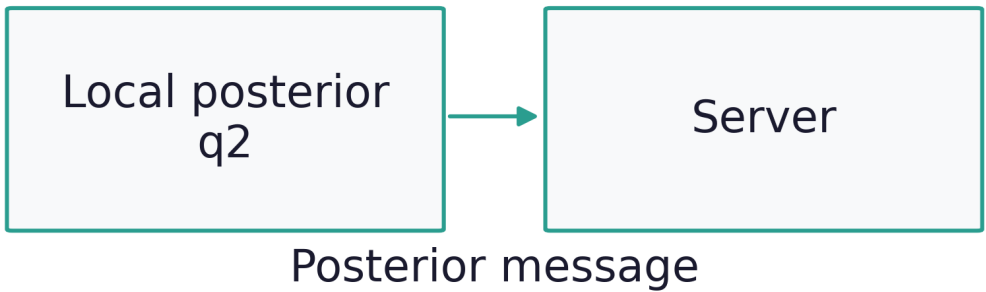
Schematic local posterior qn



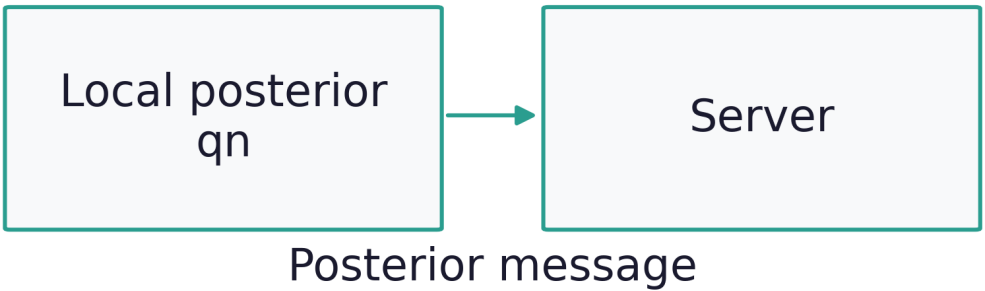
Hidden-state to action loop (part 12)



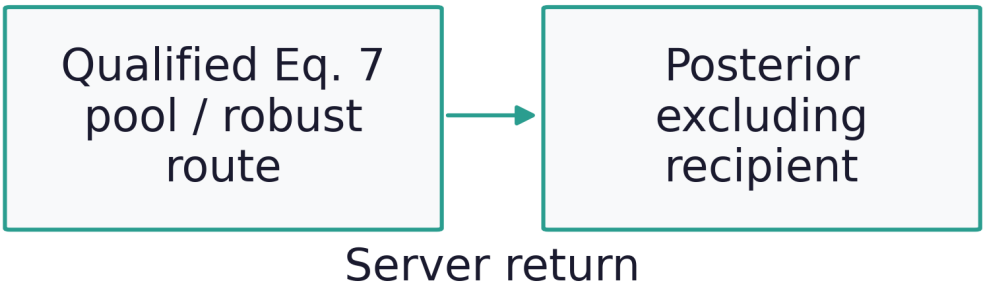
Hidden-state to action loop (part 13)



Hidden-state to action loop (part 14)

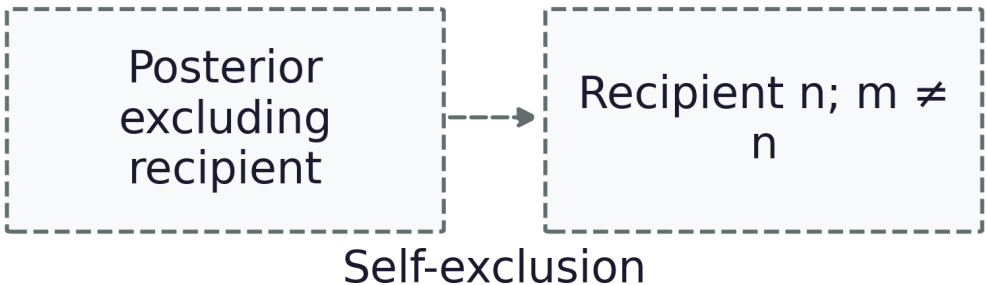


Hidden-state to action loop (part 15)

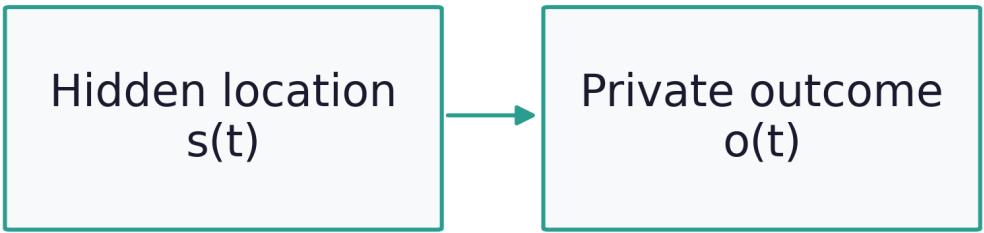


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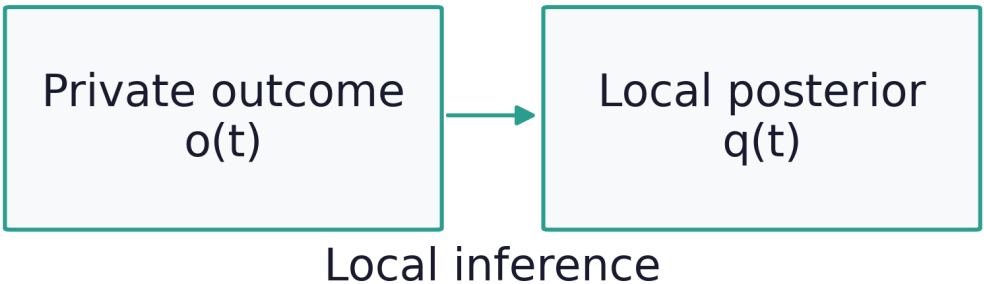
Hidden-state to action loop (part 16)



Hidden-state to action loop (part 17)

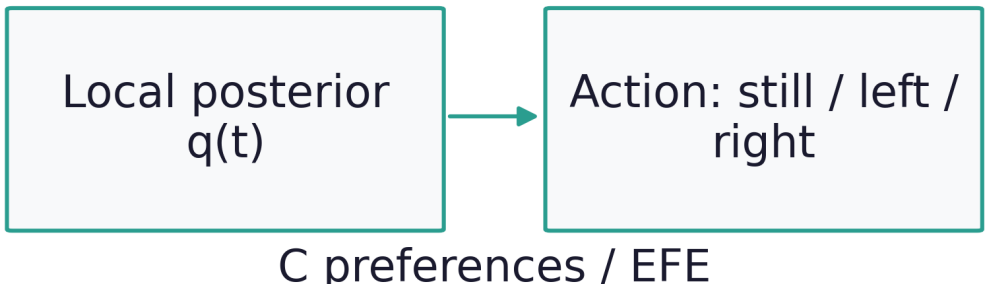


Hidden-state to action loop (part 18)



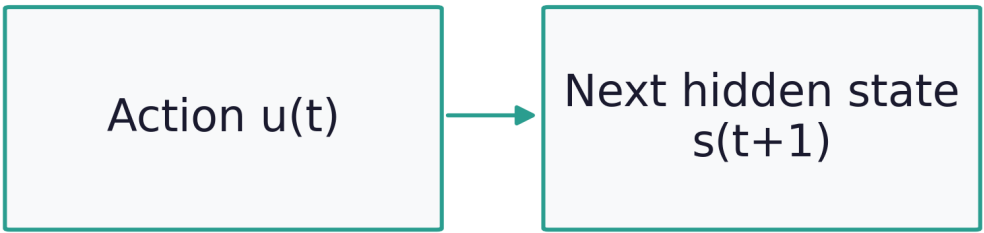
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Hidden-state to action loop (part 19)

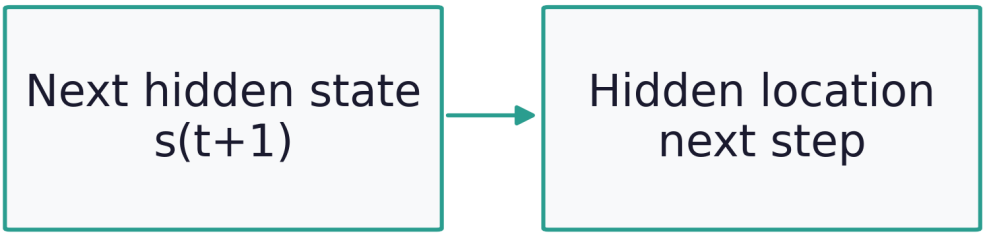


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Hidden-state to action loop (part 20)



Hidden-state to action loop (part 21)



Hidden-state to action loop (part 22)

Agents share beliefs about one hidden location. Private outcomes and controls remain local; no raw sensory data are pooled.

Hidden-state to action loop (part 23)

Flat federation uses inference and communication. The moving-world extension also executes B and EFE-guided control.

Hidden-state to action loop (part 24)

Model schematic, not an empirical result. Hatched cells only illustrate topology or posterior emphasis; they are not measured probability values.